seekdb Hybrid Retrieval Field Notes

Author: ResearchFlow demo corpus. Scope: seekdb v1.4.0.0 on OceanBase 4.3.5.3.

Page 1 records the baseline environment used for every measurement in this document. The server reports VERSION() as 5.7.25-OceanBase seekdb-v1.4.0.0 and listens on port 2881. The database used for all examples is named test.

Baseline latency for a single round trip SELECT VERSION() query measured from the local machine was 142 milliseconds including connection setup.

Page 2 - Vector index configuration

seekdb stores dense vectors in a VECTOR(n) column. The demo corpus uses n = 384, matching the all-MiniLM-L6-v2 ONNX embedding shipped inside the official pyseekdb SDK.

The index type is HNSW and the distance metric is cosine. The DDL fragment generated by pyseekdb is: WITH (DISTANCE=cosine, TYPE=hnsw, LIB=vsag).

Approximate nearest neighbour search uses this ordering clause: ORDER BY l2_distance(embedding, :query_vector) APPROXIMATE LIMIT 10 PARAMETERS(ef_search=64).

Measured cosine distance for the query 'hybrid retrieval vector and full text' against the best matching row was 0.3753, giving a similarity of 0.6247.

Page 3 - Full text analyzers

seekdb supports several FULLTEXT analyzers. The three verified in this project are ik, ngram and space. The DDL is:

```
CREATE FULLTEXT INDEX ix_name ON tbl (col) WITH PARSER ik
```

For the Chinese needle '混合召回' the ik analyzer ranked the row that actually contains that phrase first. The space analyzer ranked a different row first because it does not segment Chinese text at all.

Conclusion recorded on page 3: use the ik analyzer as the default for a bilingual Chinese and English research corpus, and keep the analyzer name configurable so ngram can be substituted without a code change.

Hybrid retrieval in seekdb fuses a full text leg and a vector leg with Reciprocal Rank Fusion. The score of a document is the sum over both legs of 1 / (rank_constant + rank), with rank_constant = 60 by default.

A row placed first in both legs scores 1/61 + 1/61 = 0.0328. A row placed first in one leg and second in the other scores 1/61 + 1/62 = 0.0325. A row present in only one leg at rank 1 scores 1/61 = 0.0164.

Those three values - 0.0328, 0.0325 and 0.0164 - are exactly the distances returned by collection.hybrid_search with rank={'rrf': {}} on the probe corpus, which is how the RRF implementation was confirmed end to end.